

Introduction to Amazon Web Services

AWS Technical Essentials — Module 1 of 7

🖥️ What Is Cloud Computing?

On-demand delivery of IT resources over the internet with **pay-as-you-go** pricing.

- No physical data centers to manage
- Provision resources in minutes
- Scale up or down on demand
- Pay only for what you consume

1. **Trade CapEx for variable expense**
2. **Massive economies of scale**
3. **Stop guessing capacity**
4. **Increase speed and agility**
5. **Stop running data centers**
6. **Go global in minutes**

🌍 AWS Global Infrastructure

🌏 Regions

- 33+ geographic locations
- Physically isolated
- Data stays in-Region

🏧 Availability Zones

- 105+ AZs globally
- One or more data centers
- Low-latency interconnects

⚡ Edge Locations

- 600+ Points of Presence
- CloudFront CDN caching
- Route 53 DNS hosting

🔒 Shared Responsibility Model

AWS Responsibility

"Security OF the cloud"

- Physical data centers
- Hardware and networking
- Hypervisor and host OS

Your Responsibility

"Security IN the cloud"

- Data and encryption
- IAM and access control
- OS patches and firewalls

AWS Identity and Access Management

IAM controls *who* can access *what* in your AWS account.

- Global service (not Region-specific)
- Free to use, no additional charges
- Principle of least privilege
- Supports MFA for extra security

👥 Users, Roles, and Policies

👤 Users

- Represent a person or app
- Long-term credentials
- Console or programmatic

🎭 Roles

- Temporary credentials
- Assumed by services or users
- Preferred for EC2 apps

📜 Policies

- JSON permission documents
- Attach to users or roles
- Allow or deny actions

🛠️ Lab Time

Introduction to AWS IAM

⏱️ 20 minutes

- Explore IAM users, groups, and roles
- Attach policies and test permissions
- Understand the effect of least privilege